Calculus I

Tutorial #11 (09–13 Dec 2024)

Topics:

- Sequences
- Infinite Series
- Integral Test; Comparison Test
- Ratio Test and Root Test

Which of the sequences a_n in the following exercises converge, and which diverge? Find the limit of each convergent sequence.

Problem 1.

(a) $a_n = \sqrt[n]{4^n n}$.

(b) $a_n = \left(\frac{1}{3}\right)^n + \frac{1}{\sqrt{2^n}}$.

Problem 2. Let (a_n) be defined as follows

$$a_1 = -1, \quad a_{n+1} = \frac{a_n + 6}{a_n + 2}.$$

Assume that the sequence converges. Find its limit.

Problem 3. Find a formula for the n-th partial sum of each series and use it to find the series sum if the series converges.

$$\frac{5}{1 \cdot 2} + \frac{5}{2 \cdot 3} + \frac{5}{3 \cdot 4} + \cdots + \frac{5}{n(n+1)} + \cdots$$

Problem 4. Express each of the numbers in the following exercise as the ratio of two integers.

$$0.\overline{23} = 0.23\,23\,23\cdots$$

Problem 5. Which series converge, and which diverge? If a series converges, find its sum. Justify your answers.

(a) $\sum_{n=0}^{\infty} \cos\left(\frac{n\pi}{2}\right)$.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^n}{n!}$.

Problem 6. *Logarithmic p-series*

(a) Show that the improper integral

$$\int_2^{\infty} \frac{dx}{x(\ln x)^p} \quad (p \text{ is a positive constant})$$

converges if and only if $p > 1$.

(b) What implications does the fact in part (a) have for the convergence of the series

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}?$$

Justify your answer.

Problem 7. Use the Integral Test to determine if a series converges or diverges.

(a) $\displaystyle\sum_{n=1}^{\infty} ne^{-n^2}$.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^{\ln n}}$.

Problem 8. Use the Comparison Test to determine if a series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}.$$

Problem 9. Use the Limit Comparison Test to determine if a series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{5^n}{\sqrt{n}4^n}.$$

Problem 10. Use any method to determine if the series converges or diverges. Justify your answer.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{10^n}$.

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^n}{(2^n)^2}$.

THE END

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Answer to selected problems

1. (a) 4 (b) 0
2. 2
3. $S = 5$
4. $\dfrac{23}{99}$
5. (a) diverges (b) diverges
7. (a) $\dfrac{1}{2e}$ (b) ∞
8. converges
9. diverges
10. (a) diverges (b) diverges